

Transcendence of π : Post-Talk Debrief

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December 2, 2025

§1 The Talk

I gave a talk proving the transcendence of π recently (Nov 4, G1 seminar). The talk went as follows:

1. Established the main steps to any transcendence proof; that is, constructing some X that is
 - (a) nonzero
 - (b) small (for analytic reasons)
 - (c) discretized away from 0 (assuming algebraicity)
2. Gave a couple simple examples (e.g. irrationality of e)
3. Overviewed big results, mentioning that the modern approach is fucked up and that a sort of master theorem is that we can control simultaneous algebraic outputs of meromorphic functions satisfying diffeqs.
4. Proved π is irrational, using (and hence motivating) integrals.
5. Presented the standard proof of Lindemann-Weierstrass, thus recovering the irrationality of e .

I now have some disagreements with the way I presented this talk, and want to record how i would have done it differently.

§2 How I'd give the talk today

The first part, through the examples of the basic proof structure, is good.

§2.1 Lindemann-Weierstrass

I think the correct next step is to skip all the foreplay and write down the following equation:

$$0.0034963 \approx \int_0^3 \frac{x^{10}(3-x)^{10}}{10!} e^{3-x} dx = 81(1741039699e^3 - 34969717159). \quad (1)$$

Let the audience contemplate this equation. Enough contemplation establishes that

1. Taking $N = 10 \rightarrow \infty$ proves e^3 is irrational (since $N! \gg C^n$)
2. The reason that this proof is even plausible is that $(e^x)' = e^x$. Roughly, e^x satisfying some diffeq means the integration by parts doesn't yield nonsense.

Theres some actual work to prove the full Lindemann-Weierstrass from here, but this gets the main punch through efficiently. I also made the choice to actually emphasize the used special property of e^x , because I've realized it ties into the modern approach to transcendence.

§2.2 Schneider-Lang

I will break the style of a talk outline now because I want to write down a probably impractical number of details.

We present a proof of the following theorem:

Theorem 1 (Schneider-Lang). *Let K be a number field, and let $K[f_1, f_2, \dots, f_t]$ be a set of meromorphic functions closed under differentiation. Suppose f_1, f_2 are algebraically independent with $f_i(z) \ll e^{|z|^{\rho_i}}$. Then, there are at most $(\rho_1 + \rho_2)[K : \mathbb{Q}]$ values of z such that $f_i(z) \in K$ for all i .*

I will note that it is common practice in presentations of Schneider-Lang to prove some weaker bound like $20\rho[K : \mathbb{Q}]$, but I want to actually track an exact bound because I believe it's enlightening.

Proof. Let d denote $[K : \mathbb{Q}]$. Suppose the f_i are simultaneously in K at $\omega_1, \dots, \omega_m$. We will show that $m \leq (\rho_1 + \rho_2)d$ by constructing a function

$$X(z) = \sum_{i,j} c_{i,j} f_i^j(z) g^j(z) \quad (2)$$

such that

1. X vanishes to order N at each ω_i .
2. Some higher order derivative (WLOG this is $X^{(N+1)}(\omega_1)$) is nonvanishing (from algebraic independence!)
3. The coefficients are negligible (below $(N!)^\varepsilon$)
4. We have $|X^{(N+1)}(\omega_1)| \ll (N!)^{1-m/(\rho_1+\rho_2)}$; this is complex analysis exploiting the high vanishing at m points.
5. We have $|X^{(N+1)}(\omega_1)| \gg \frac{1}{(N!)^{d-1}}$; this is discretization from being algebraic with small conjugates.

When we write $A \ll B$, we can be ignoring up to a factor of C^N .

Remark. *Note the resemblance with our integration from earlier! Recall that $f(z) = z^N(3-z)^N e^{-z}$ vanished to high order at 0 and 3, and note that integration by parts roughly gives us higher derivatives of f . So this is in line with our old proof style. The strange thing to process is that the analysis step is weird now; we're somehow going to exploit the number of algebraic points?*

Suppose we sum over $i \leq 2\varepsilon^{-1/2}m^{1/2}N^{\rho_2/(\rho_1+\rho_2)}$ and $j \leq 2\varepsilon^{-1/2}m^{1/2}N^{\rho_1/(\rho_1+\rho_2)}$ so a $4\varepsilon^{-1}$ factor over the number of needed free variables chosen to balance the growth rates. Siegel's lemma then tells us that we may pick coefficients such that

$$B = \max c_{i,j} \ll (N!)^\varepsilon \quad (3)$$

Imagining the algebraic conjugates of $X^{(N+1)}(\omega_1)$, one sees that they're $\ll BN! \cdot C^N$, so Vieta's tells us that

$$|X^{(N+1)}(\omega_1)| \gg \frac{1}{(BN! \cdot C^N)^{d-1}}. \quad (4)$$

On the other hand, one may observe that by the maximum modulus principle,

$$\left| \frac{X^{N+1}(\omega_1)}{N!} \right| \leq \sup_{|z|=R} \frac{X(z)C^N}{\prod (z - \omega_i)^N} \ll B \cdot \frac{e^{R^{\rho_1} N^{\rho_2/(\rho_1+\rho_2)} + R^{\rho_2} N^{\rho_1/(\rho_1+\rho_2)}}}{R^{mN}}. \quad (5)$$

Taking the visually appealing $R = N^{1/(\rho_1+\rho_2)}$ gives

$$|X^{N+1}(\omega_1)| \ll \frac{N!}{(N!)^{m/(\rho_1+\rho_2)}}. \quad (6)$$

Comparing Eq. (4) and Eq. (6), we are done. $\square$

§2.3 Comprehending Schneider-Lang

Going back into a sort of talk voice now. How does one present this in relationship to Lindemann-Weierstrass? The standard way of proving numbers are irrational with Schneider-Lang is to show not only that they are outputs from meromorphic functions satisfying a $\mathbb{Q}$ differential equation, but to bootstrap them to infinitely many outputs. For example, e is transcendental, since otherwise all n would be a simultaneous small degree algebraic output of x and e^x . I think that giving a more finitary presentation for Lindemann-Weierstrass can be enlightening.

Suppose e is algebraic with degree d . Then, we have $1, e, \dots, e^d \in K$, where K has degree d . Tracing through the proof of Schneider-Lang with the above inputs, we find that we are analyzing sums of the form $\sum_{i,j} a_{i,j} x^i e^{jx}$ with $i \ll N$ and $j \ll N^\varepsilon$ with high vanishing at $0, 1, \dots, d$.

Compare this with Lindemann-Weierstrass, where we explicitly construct $e^{-x} (\prod (x - \alpha))^N$. Somehow we bypass the need to use the discretization by just using a couple more e terms instead. Honestly makes me mad with how barely it escapes having a tangible case; going through the proof just analyzing $e^{-x} (\prod (x - \alpha))^N$ verifies that it analogously barely fails (without the extra discretization).